

Enterprise Fraud Detection Intelligence System

Final Internship Project Report
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Project Objective

The objective of this project was to develop an end-to-end enterprise fraud detection intelligence system capable of identifying high-risk financial transactions using machine learning, Explainable AI, and operational risk segmentation.

Dataset Overview

The project utilized the IEEE-CIS Fraud Detection dataset from Kaggle. The train_transaction.csv and train_identity.csv files were merged using the TransactionID column to build a unified fraud intelligence dataset containing over 590,000 transactions and 434 features.

Data Preprocessing & Analysis

Data preprocessing included memory optimization, missing value analysis, feature engineering, categorical encoding, and dataset cleaning. Exploratory Data Analysis identified strong fraud imbalance with approximately 3.5% fraudulent transactions.

Key Exploratory Insights

- Product category C demonstrated elevated fraud exposure.
- Fraudulent transactions exhibited distinct transaction amount behavior.
- Several identity-related features contained high missing value percentages.
- Transaction behavioral variables significantly influenced fraud probability.

Machine Learning Models

Multiple machine learning models were implemented and evaluated including Logistic Regression, Random Forest, and XGBoost. Performance evaluation was conducted using Accuracy, Precision, Recall, F1-Score, and ROC-AUC metrics.

Model Performance Summary

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.741	0.192	0.331	0.243	0.601
Random Forest	0.924	0.882	0.458	0.603	0.889
XGBoost	0.922	0.858	0.455	0.594	0.897

Best Performing Model

XGBoost achieved the strongest fraud detection capability with the highest ROC-AUC score and effective fraud discrimination performance. The model successfully captured nonlinear transaction behavior patterns and complex fraud relationships.

Explainable AI using SHAP

SHAP (SHapley Additive exPlanations) was implemented to improve model transparency and identify the most influential fraud-driving features. Key fraud indicators included V257, V258, transaction amount behavior, and email domain intelligence.

Risk Segmentation Framework

Transactions were segmented into Low Risk, Medium Risk, and High Risk categories. High-risk transactions demonstrated extremely elevated fraud probability, enabling operational prioritization for fraud investigation teams.

Interactive Dashboard

A professional Streamlit-based dashboard was developed to visualize fraud analytics, model performance, Explainable AI insights, and operational fraud intelligence. The dashboard also includes an interactive fraud risk simulator.

Business Impact

This system demonstrates how machine learning and Explainable AI can support enterprise fraud prevention, financial risk reduction, operational monitoring, intelligent investigation prioritization, and analyst decision support systems.

Conclusion

This project successfully demonstrates an enterprise-level fraud intelligence pipeline integrating machine learning, Explainable AI, operational analytics, and interactive dashboarding. The final system provides accurate fraud detection, interpretable predictions, and actionable fraud intelligence insights suitable for real-world financial fraud analysis workflows.